

Differential Geometry

Exercise 10

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10.1 Problem (DoCarmo, Riemannian Geometry, page 103 question 1)

Let G be a Lie group with a bi-invariant metric. Let $X, Y, Z \in \mathfrak{g}$ be left-invariant vector fields.

- (1) Show that $\nabla_X Y = \frac{1}{2}[X, Y]$.
- (2) Conclude that $R(X, Y)Z = \frac{1}{4}[[X, Y], Z]$.
- (3) Show that if X, Y are orthonormal, and σ is the plane generated by X, Y , then the sectional curvature of G at σ is

$$K(\sigma) = \frac{1}{2} |[X, Y]|^2.$$

- (1) Consider the map $\mathfrak{g}^2 \rightarrow \mathfrak{g}$ by $(X, Y) \mapsto \nabla_X Y$. This is $\mathbb{R}$ -bilinear, as we know this of connections, and further we know that $\nabla_X X = 0$. This implies that it is antisymmetric; a bilinear map $\alpha: V^2 \rightarrow W$ which is zero when its inputs are equal is antisymmetric:

$$0 = \alpha(v + u, v + u) = \alpha(v, v) + \alpha(v, u) + \alpha(u, v) + \alpha(u, u) = \alpha(v, u) + \alpha(u, v).$$

Thus $\nabla_X Y = -\nabla_Y X$ and by symmetry of the connection,

$$[X, Y] = \nabla_X Y - \nabla_Y X = 2\nabla_X Y$$

as desired.

- (2) By the previous point,

$$R(X, Y)Z = \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X, Y]} Z = \frac{1}{4}[Y, [X, Z]] - \frac{1}{4}[X, [Y, Z]] + \frac{1}{2}[[X, Y], Z].$$

Now by Jacobi,

$$0 = [X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = [X, [Y, Z]] + [Z, [X, Y]] - [Y, [X, Z]]$$

which implies

$$[Y, [X, Z]] - [X, [Y, Z]] = [Z, [X, Y]] = -[[X, Y], Z].$$

Thus we have

$$R(X, Y)Z = \frac{1}{4}[[X, Y], Z]$$

as desired.

- (3) If X, Y are orthonormal then the sectional curvature of σ is just $\text{Rm}(X, Y, X, Y)$ which is

$$K(\sigma) = \langle R(X, Y)X, Y \rangle = \frac{1}{4} \langle [[X, Y], X], Y \rangle = -\frac{1}{4} \langle \nabla_X [X, Y], Y \rangle.$$

Now, because the connection is metric,

$$X \langle [X, Y], Y \rangle = \langle \nabla_X [X, Y], Y \rangle + \langle [X, Y], \nabla_X Y \rangle.$$

And similarly

$$0 = X\langle Y, Y \rangle = 2\langle \nabla_X Y, Y \rangle,$$

where the left-hand side is zero since $\langle Y, Y \rangle = 1$ is constant. Thus $0 = 2\langle \nabla_X Y, Y \rangle = \langle [X, Y], Y \rangle$, and so we see

$$0 = \langle \nabla_X [X, Y], Y \rangle + \langle [X, Y], \nabla_X Y \rangle \implies \langle [X, [X, Y]], Y \rangle = -\langle [X, Y], [X, Y] \rangle,$$

and finally

$$K(\sigma) = \frac{1}{4} \langle [X, Y], [X, Y] \rangle$$

as desired.

10.2 Problem

- (1) Let ∇ be a connection on a manifold M . For vector fields $X, Y, Z \in \mathfrak{X}(M)$, let $\nabla^2 Z$ be the $(2, 1)$ tensor field obtained by taking the covariant derivative of the $(1, 1)$ tensor field ∇Z . We write $\nabla_{X,Y}^2 Z$ for $\nabla^2 Z(X, Y)$. Show that

$$R(X, Y)Z = \nabla_{Y,X}^2 Z - \nabla_{X,Y}^2 Z.$$

- (2) (DoCarmo, Riemmanian Geometry, page 106 problem 7): prove the second **Binachi identity**:

$$\nabla \text{Rm}(X, Y, Z, W, T) + \nabla \text{Rm}(X, Y, W, T, Z) + \nabla \text{Rm}(X, Y, T, Z, W) = 0.$$

- (1) By the formula for the covariant derivative of a tensor, we see that for a covector field ω ,

$$\begin{aligned} \nabla_{Y,X}^2 Z(\omega) &= \nabla_Y \nabla Z(\omega, X) = Y(\nabla Z(\omega, X)) - \nabla Z(\nabla_Y \omega, X) - \nabla Z(\omega, \nabla_Y X) \\ &= Y(\omega(\nabla_X Z)) - \nabla_Y \omega(\nabla_X Z) - \omega(\nabla_{\nabla_Y X} Z). \end{aligned}$$

Now we know that in general,

$$Y(\omega X) = \nabla_Y(\omega X) = \nabla_Y \omega(X) + \omega(\nabla_Y X),$$

and thus

$$Y(\omega(\nabla_X Z)) = \nabla_Y \omega(\nabla_X Z) + \omega(\nabla_Y \nabla_X Z).$$

So

$$\nabla_{Y,X}^2 Z(\omega) = \omega(\nabla_Y \nabla_X Z) - \omega(\nabla_{\nabla_Y X} Z).$$

Since this is true for all ω , we see that

$$\nabla_{Y,X}^2 Z = \nabla_Y \nabla_X Z - \nabla_{\nabla_Y X} Z.$$

Thus

$$\begin{aligned} \nabla_{Y,X}^2 Z - \nabla_{X,Y}^2 Z &= \nabla_Y \nabla_X Z - \nabla_{\nabla_Y X} Z - \nabla_X \nabla_Y Z + \nabla_{\nabla_X Y} Z \\ &= \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{\nabla_X Y - \nabla_Y X} Z \\ &= \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X,Y]} Z \\ &= R(X, Y)Z. \end{aligned}$$

- (2) Everything here is tensorial so we can choose whatever local frame we want, show it for that, and extrapolate the result for all vector fields. Let $p \in M$ and choose a geodesic field (E_i) at p . Then we have,

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = \nabla_{E_h} \text{Rm}(E_i, E_j, E_k, E_\ell) = E_h(\text{Rm}(E_i, E_j, E_k, E_\ell)) - \text{Rm}(\nabla_{E_h} E_i, E_j, E_k, E_\ell) - \dots - \text{Rm}(E_i, \dots, \nabla_{E_h} E_\ell).$$

At p we know $\nabla_{E_i} E_j|_p = 0$ for all i, j , so by multilinearity the last terms above are all zero (at p). So we see

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = E_h \text{Rm}(E_i, E_j, E_k, E_\ell) = E_h \left\langle \nabla_{E_j, E_i}^2 E_k - \nabla_{E_i, E_j}^2 E_k, E_\ell \right\rangle = \left\langle \nabla_{E_h} (\nabla_{E_j, E_i}^2 E_k - \nabla_{E_i, E_j}^2 E_k), E_\ell \right\rangle,$$

the last equality is due to the connection being metric and $\nabla_{E_h} E_\ell = 0$ at p . Now,

$$\left\langle \nabla_{E_h} \nabla_{E_j, E_i}^2 E_k, E_\ell \right\rangle = \left\langle \nabla_{E_h} \nabla_{E_j} \nabla_{E_i} E_k - \nabla_{E_h} \nabla_{E_i} \nabla_{E_j} E_k + \nabla_{E_h} \nabla_{[E_i, E_j]} E_k, E_\ell \right\rangle,$$

and so by this and symmetry of the connection,

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = \left\langle \nabla_{E_h} \nabla_{E_j} \nabla_{E_i} E_k - \nabla_{E_h} \nabla_{E_i} \nabla_{E_j} E_k + \nabla_{E_h} \nabla_{[E_i, E_j]} E_k, E_\ell \right\rangle.$$

Notice that $\text{Rm}(E_i, E_j, E_k, E_\ell) = \text{Rm}(E_k, E_\ell, E_i, E_j)$ so we also can write

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = \left\langle \nabla_{E_h} \nabla_{E_i} \nabla_{E_k} E_j - \nabla_{E_h} \nabla_{E_k} \nabla_{E_i} E_j + \nabla_{E_h} \nabla_{[E_k, E_i]} E_j, E_\ell \right\rangle.$$

And so we have that the sum gives the inner product of the following with E_j (writing ∇_i for ∇_{E_i} and $\nabla_{i,j}$ for $\nabla_{[E_i, E_j]}$):

$$\begin{aligned} & \nabla_h \nabla_\ell \nabla_k E_i - \nabla_h \nabla_k \nabla_\ell E_i + \nabla_h \nabla_{k,\ell} E_i \\ & + \nabla_\ell \nabla_k \nabla_h E_i - \nabla_\ell \nabla_h \nabla_k E_i + \nabla_\ell \nabla_{h,k} E_i \\ & + \nabla_k \nabla_h \nabla_\ell E_i - \nabla_k \nabla_\ell \nabla_h E_i + \nabla_k \nabla_{\ell,h} E_i. \end{aligned}$$

Now consider

$$R(E_\ell, E_h) \nabla_k E_i = \nabla_h \nabla_\ell \nabla_k E_i - \nabla_\ell \nabla_h \nabla_k E_i + \nabla_{\ell,h} \nabla_k E_i.$$

These are all zero at p since $\nabla_k E_i = 0$, and the first two terms are the terms from the the above large sum. The final term is zero since $[E_\ell, E_h] = \nabla_\ell E_h - \nabla_h E_\ell = 0$, so the first two terms above cancel out, and so the large sum is just the sum of the terms of the form $\nabla_h \nabla_{k,\ell} E_i$. Also,

$$\nabla_h \nabla_{k,\ell} E_i = \nabla_h \nabla_{k,\ell} E_i - \nabla_{k,\ell} \nabla_h E_i = -R(E_h, [E_k, E_\ell]) E_i + \nabla_{[E_h, [E_k, E_\ell]]} E_i,$$

and since $[E_k, E_\ell] = 0$ at p we see that

$$\nabla_h \nabla_{k,\ell} E_i = \nabla_{[E_h, [E_k, E_\ell]]} E_i.$$

Thus the large sum is the sum of terms of this form, as h, k, ℓ cycle.

Thus the large sum is equal to

$$\sum_{(h,k,\ell)} \nabla_{[E_h, [E_k, E_\ell]]} E_i = \nabla_{\sum [E_h, [E_k, E_\ell]]} E_i,$$

where the sum is over the cycles of (h, k, ℓ) . By the Jacobi identity for the Lie bracket, this is zero. And thus we get that the sum of the covariant derivatives of Rm is the inner product of zero with E_j , which is zero.

10.3 Problem

Let $E \rightarrow M$ be a vector bundle with metric h and connection $\bar{\nabla}$. Let $S \subseteq E$ be a subbundle with orthogonal complement $S^\perp \subseteq E$ so that $E = S \oplus S^\perp$. Let $P^\perp: E \rightarrow S^\perp$ denote the orthogonal projection along S . Let $\nabla, \nabla^\perp$ denote the induced connections on $S, S^\perp$ respectively. Finally let $\Pi: TM \otimes S \rightarrow S^\perp$ denote the second fundamental form of S and $\bar{R}: \Lambda^2 TM \otimes E \rightarrow E$ denote the curvature of $\bar{\nabla}$.

- (1) Show that for $X, Y \in \mathfrak{X}(M)$ and $\xi \in \Gamma(S)$,

$$P^\perp \bar{R}(X, Y)\xi = \nabla_Y^\perp(\Pi(X)\xi) - \Pi(X)\nabla_Y\xi - \nabla_X^\perp(\Pi(Y)\xi) + \Pi(Y)\nabla_X\xi + \Pi([X, Y])\xi.$$

- (2) Let $M \subseteq \bar{M}$ be a submanifold, $\bar{g}$ a Riemannian metric on $\bar{M}$ and g the induced metric on M . Let $E = T\bar{M}|_M$ and $S = TM$ so that $S^\perp = \nu_M$. Let $\bar{\nabla}$ be the connection induced from the Levi-Civita connection of $(\bar{M}, \bar{g})$ so that ∇ is the Levi-Civita connection on (M, g) .

- (i) Define $\nabla\Pi: \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(\nu_M)$ by

$$\nabla_Y\Pi(X)Z = \nabla_Y^\perp(\Pi(X)Z) - \Pi(\nabla_Y X)Z - \Pi(X)\nabla_Y Z, \quad X, Y, Z \in \mathfrak{X}(M).$$

Show that $\nabla\Pi$ is tensorial, and thus defines a bundle morphism

$$\nabla\Pi: TM \otimes TM \otimes TM \rightarrow \nu_M.$$

- (ii) Show that

$$P^\perp \bar{R}(X, Y)Z = \nabla_Y\Pi(X)Z - \nabla_X\Pi(Y)Z.$$

- (1) The proof here is essentially the same as that of Gauss's theorem's proof from class. By definition we have for $X \in \mathfrak{X}(M)$ and $\xi \in \Gamma(S)$,

$$\bar{\nabla}_X\xi = \nabla_X\xi + \Pi(X)\xi.$$

Since $\nabla_X\xi \in \Gamma(S)$ and $\Pi(X)\xi \in \Gamma(S^\perp)$ repeating this we see

$$\bar{\nabla}_Y\bar{\nabla}_X\xi = \nabla_Y\nabla_X\xi + \Pi(Y)\nabla_X\xi + \nabla_Y^\perp\Pi(X)\xi + \Pi^\perp(Y)\Pi(X)\xi.$$

The first and last terms here lie in $\Gamma(S)$, while the middle two lie in $\Gamma(S^\perp)$. Thus projecting along S gives

$$P^\perp\bar{\nabla}_Y\bar{\nabla}_X\xi = \Pi(Y)\nabla_X\xi + \nabla_Y^\perp\Pi(X)\xi.$$

And for the same reason as before, by definition,

$$\bar{\nabla}_{[X, Y]}\xi = \nabla_{[X, Y]}\xi + \Pi([X, Y])\xi,$$

and here the first term is in $\Gamma(S)$ while the second is in $\Gamma(S^\perp)$ so that

$$P^\perp\bar{\nabla}_{[X, Y]}\xi = \Pi([X, Y])\xi.$$

Putting this all together,

$$\begin{aligned} P^\perp\bar{R}(X, Y)\xi &= P^\perp(\bar{\nabla}_Y\bar{\nabla}_X\xi - \bar{\nabla}_X\bar{\nabla}_Y\xi + \bar{\nabla}_{[X, Y]}\xi) \\ &= \Pi(Y)\nabla_X\xi + \nabla_Y^\perp\Pi(X)\xi - \Pi(X)\nabla_Y\xi - \nabla_X^\perp\Pi(Y)\xi + \Pi([X, Y])\xi, \end{aligned}$$

which is the desired expression.

(2)

- (i) By additivity of $\Pi, \nabla, \nabla^\perp$ in all parameters, $\nabla\Pi$ is clearly additive. And since Π is tensorial and $\nabla, \nabla^\perp$ are tensorial in their lower parameters, $\nabla\Pi$ is tensorial in Y . It remains to show that $\nabla\Pi$ is multiplicative in X, Z :

$$\nabla_Y\Pi(fX)Z = f\nabla_Y\Pi(X)Z, \quad \nabla_Y\Pi(X)(fZ) = f\nabla_Y\Pi(X)Z \quad \text{for } f \in C^\infty(M).$$

For X , we see

$$\begin{aligned} \nabla_Y\Pi(fX)Z &= \nabla_Y^\perp(\Pi(fX)Z) - \Pi(\nabla_Y fX)Z - \Pi(fX)\nabla_Y Z \\ &= \nabla_Y^\perp(f\Pi(X)Z) - \Pi((Yf)X + f\nabla_Y X)Z - f\Pi(X)\nabla_Y Z \\ &= (Yf)\Pi(X)Z + f\nabla_Y^\perp(\Pi(X)Z) - (Yf)\Pi(X)Z - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z \\ &= f(\nabla_Y^\perp(\Pi(X)Z) - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z) \\ &= f\nabla_Y\Pi(X)Z. \end{aligned}$$

And similarly for Z , we see

$$\begin{aligned} \nabla_Y\Pi(X)(fZ) &= \nabla_Y^\perp(\Pi(X)(fZ)) - \Pi(\nabla_Y X)(fZ) - \Pi(X)\nabla_Y(fZ) \\ &= \nabla_Y^\perp(f\Pi(X)Z) - f\Pi(\nabla_Y X)Z - \Pi(X)((Yf)Z + f\nabla_Y Z) \\ &= (Yf)\Pi(X)Z + f\nabla_Y^\perp(\Pi(X)Z) - f\Pi(\nabla_Y X)Z - (Yf)\Pi(X)Z - f\Pi(X)\nabla_Y Z \\ &= f(\nabla_Y^\perp(\Pi(X)Z) - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z) \\ &= f\nabla_Y\Pi(X)Z. \end{aligned}$$

So that $\nabla\Pi$ is tensorial (linear) in all parameters.

- (ii) From our first point, it is sufficient to show that

$$\nabla_Y\Pi(X)Z - \nabla_X\Pi(Y)Z = \nabla_Y^\perp(\Pi(X)Z) - \Pi(X)\nabla_Y Z - \nabla_X^\perp(\Pi(Y)Z) + \Pi(Y)\nabla_X Z + \Pi([X, Y])Z.$$

Direct computation yields

$$\nabla_Y\Pi(X)Z - \nabla_X\Pi(Y)Z = \nabla_Y^\perp(\Pi(X)Z) - \Pi(\nabla_Y X)Z - \Pi(X)\nabla_Y Z - \nabla_X^\perp(\Pi(Y)Z) + \Pi(\nabla_X Y)Z + \Pi(Y)\nabla_X Z.$$

Cancelling out similar terms we see that we need

$$\Pi(\nabla_X Y)Z - \Pi(\nabla_Y X)Z = \Pi([X, Y])Z.$$

And indeed, by tensorality of Π the left-hand side is equal to

$$\Pi(\nabla_X Y - \nabla_Y X)Z = \Pi([X, Y])Z,$$

where the equality is due to symmetry of the Levi-Civita connection.

10.4 Problem (Do Carmo, Differential Geometry of Curves and Surfaces, q7 p169)

Consider the surface of revolution $(\varphi(v)\cos u, \varphi(v)\sin u, \psi(v))$ with constant Gaussian curvature

K . Parameterize φ, ψ such that $(\varphi')^2 + (\psi')^2 = 1$.

- (1) Show that φ satisfies $\varphi'' + K\varphi = 0$ and ψ is given by $\psi = \int \sqrt{1 - (\varphi')^2}$.
- (2) Show that all surfaces of revolution with constant curvature $K = 1$ which intersect the xy plane orthogonally are given by

$$\varphi(v) = C \cos v, \quad \psi(v) = \int_0^v \sqrt{1 - C^2 \sin^2 v}.$$

for some constant C .

- (3) Determine all surfaces of revolution with constant curvature $K = -1$, and show that they are all of one of the following forms

$$\varphi(v) = e^v, \quad \varphi(v) = \cosh(v), \quad \varphi(v) = \sinh(v).$$

- (4) Show that the only surface of revolution with $K = 0$ are the right cylinder, the right circular cone, and the plane.

- (1) We know that (Do Carmo page 162)

$$K = -\frac{\varphi''}{\varphi},$$

and thus $\varphi'' + K\varphi = 0$. And we know

$$(\psi')^2 = 1 - (\varphi')^2,$$

we take the square root and integrate, giving us both equations.

- (2) Here we need to solve the ODE $\varphi'' + \varphi = 0$. The characteristic polynomial of this ODE is $x^2 + 1 = 0$, which has roots $\pm i$. Taking the real and imaginary parts, we have solutions $\cos(v), \sin(v)$ and so the general solution is given by $A \cos(v) + B \sin(v)$.

In order for the surface of revolution to intersect xy orthogonally, all meridians must attain critical points when $v = 0$. This is because geometrically, the meridian should be flat when intersecting the xy plane, i.e. have a critical point. This just means $\varphi'(0) = 0$; so $0 = -A \sin(0) + B \cos(0) = B$.

So the surfaces of revolution are all of the form $\varphi(v) = A \cos(v)$ and since $\varphi'(v) = -A \sin(v)$,

$$\varphi(v) = A \cos(v), \quad \psi(v) = \int_0^v \sqrt{1 - A^2 \sin^2(v)}.$$

- (3) We need to solve $\varphi'' - \varphi = 0$, which has characteristic polynomial $x^2 - 1$. The roots are $x = \pm 1$, which means φ has general form $\varphi(v) = Ae^v + Be^{-v}$. If $B = 0$ we are finished, if $A = 0$ then we can reparameterize v as $-v$ to get $\varphi(v) = Be^v$. Otherwise we claim that a shift of v will give us $\varphi(v) = C \cosh(v)$ or $C \sinh(v)$. So reparameterize by $v + c$, then

$$\varphi(v) = Ae^{v+c} + Be^{-v-c} = Ae^c \cdot e^v + Be^{-c} \cdot e^{-v}.$$

So we want to find a c such that

$$|Ae^c| = |Be^{-c}| \iff e^{2c} = \left| \frac{B}{A} \right| \iff c = \frac{1}{2} \log \left| \frac{B}{A} \right|.$$

In such a case, letting $2C = Ae^c$ we see that

$$\varphi(v) = Ce^v \pm Ce^{-v},$$

which is $C \cosh(v)$ or $C \sinh(v)$ (depending on sign), as desired.

The formula for ψ is due to $\cosh'(v) = \sinh(v)$ and $\sinh'(v) = -\cosh(v)$ and $(e^v)' = e^v$.

- (4) We need to solve $\varphi'' = 0$, which has solution $\varphi(v) = Av + B$. Then we have $\psi(v) = Cv$ for $C = \sqrt{1 - A^2}$.

When $A = 0$ this gives us

$$(B \cos u, B \sin u, v)$$

which is the cylinder. When $A = 1$ we have

$$((v + B) \cos u, (v + B) \sin u, 0),$$

which is the plane; we can shift v to $v + B$ so this is just polar coordinates of the plane. In general for nonzero A we can shift v so that $B = 0$.

So for all other A , we have

$$(Av \cos u, Av \sin u, Cv),$$

and since $A \neq 1$, $C \neq 0$. This is the cone, as it is the stretching of $(v \cos u, v \sin u, v)$ which is the classic parameterization of the cone.

10.5 Problem

- (1) Show that any smooth manifold M admits a Riemannian metric.

- (2) Let ∇ be the Levi-Civita connection on M . Define, for $\omega \in \Omega^k(M)$,

$$(d\omega)(X_1, \dots, X_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} (\nabla_{X_i} \omega)(X_1, \dots, \widehat{X}_i, \dots, X_{k+1}).$$

Show that this defines a linear operator

$$d: \Omega^k(M) \rightarrow \Omega^{k+1}(M).$$

- (3) Show that in any coordinate frame, if

$$\omega = \sum_I \omega_I dx^I \quad \text{summing over increasing multi-indices,}$$

then

$$d\omega = \sum_I d\omega_I \wedge dx^I.$$

- (4) Conclude that d does not depend on the underlying Riemannian metric, and that d coincides with the construction of the exterior derivative from class.

- (5) Prove that

$$\begin{aligned} (d\omega)(X_1, \dots, X_{k+1}) &= \sum_{i=1}^{k+1} (-1)^{i+1} X_i(\omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1})) \\ &\quad + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_{k+1}). \end{aligned}$$

- (1) We can embed M in $\mathbb{R}^n$ for some n (Whitney), and then pull back the Euclidean metric to M . Alternatively, cover M by smooth charts (U_i, φ_i) where $U_i \subseteq M$ are open and $\varphi_i: U_i \rightarrow \mathbb{R}^n$ are diffeomorphisms onto open sets. For each i , pull back the Euclidean metric on $\mathbb{R}^n$ to U_i via φ_i to give $g_i = \varphi_i^* \bar{g}$. This is a metric on U_i as φ_i is a diffeomorphism.

Let (ψ_i) be a partition of unity subordinate to (U_i) , and define on all of M

$$g = \sum_i \psi_i g_i.$$

Explicitly, we extend g_i to be zero outside of U_i , and we define for $p \in M$, $g_p = \sum_i \psi_i(p) g_i|_p$. Around every $p \in M$ there is a neighborhood for which this sum is finite, and thus a finite sum of smooth tensor fields, so g itself is a smooth tensor field. Furthermore it is a metric: g_p is a linear combination of metrics $\psi_i(p) g_i|_p$ and is thus bilinear and symmetric. It is also nonnegative since each g_i and ψ_i are. And $g_p(v, v) = 0$ iff $g_i|_p(v, v) = 0$ for all i where $p \in U_i$ and $\psi_i(p) > 0$. There must be at least one such i , and since g_i is a metric this means $v = 0$, so g is positive definite.

- (2) We show note that $d\omega$ is a tensor. We can write

$$d\omega(X_1, \dots, X_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} \nabla \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}, X_i),$$

so

$$d\omega = \sum \text{sgn} \tau^\tau \nabla \omega,$$

where the sum is over all permutations τ which take a specific index i and move it to the end (meaning i is mapped to $k+1$, and $j > i$ is shifted back one). Since $\nabla \omega$ is a tensor, permuting it is a tensor, and thus $d\omega$ is a tensor.

Now we need to show that $d\omega$ is alternating. While $\nabla \omega$ itself is not alternating, $\nabla_X \omega$ is. This is because for any permutation σ ,

$$\begin{aligned} \nabla_X \omega(Y_{\sigma 1}, \dots, Y_{\sigma k}) &= X(\omega(\bar{Y}_\sigma)) - \sum_i \omega(Y_{\sigma 1}, \dots, \nabla_X Y_{\sigma i}, \dots, Y_{\sigma k}) \\ &= \text{sgn} \sigma X(\omega(\bar{Y})) - \text{sgn} \sigma \sum_i \omega(Y_1, \dots, \nabla_X Y_i, \dots, Y_k) \\ &= \text{sgn} \sigma \nabla_X \omega(\bar{Y}). \end{aligned}$$

Now, if $\bar{X}$ has a repeated value, say $X = X_j = X_{j+1}$,

$$\begin{aligned} d\omega(\bar{X}) &= \sum_i (-1)^{i+1} \nabla_{X_i} \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}) = \\ &(-1)^{j+1} \nabla_X \omega(\dots, \hat{X}_j, X, \dots) + (-1)^j \nabla_X \omega(\dots, X, \hat{X}_{j+1}, \dots), \end{aligned}$$

since all other terms in the sum have repeated inputs, and $\nabla_{X_i} \omega$ is alternating. Notice that there is an equality $(\dots, \hat{X}_j, X, \dots) = (\dots, X, \hat{X}_{j+1}, \dots)$ and so this sum is zero. A tensor which is zero when it is input an adjacent repeated value is alternating. This is because then swapping adjacent values alters the sign: say α is multilinear and zero when adjacent values are repeated, then

$$0 = \alpha(\dots, X+Y, X+Y, \dots) = \alpha(\dots, X, X, \dots) + \alpha(\dots, X, Y, \dots) + \alpha(\dots, Y, X, \dots) + \alpha(\dots, Y, Y, \dots) = \alpha(\dots, X, Y, \dots) + \alpha(\dots, Y, X, \dots).$$

Since permutations of the form $(i, i+1)$ generate S_n , this means that such an α is then alternating.

So $d\omega$ is alternating, meaning is in $\Omega^{k+1}(M)$ as desired. Clearly by linearity of $\nabla \omega$ in ω , d is linear.

- (3) Let us denote by D the linear map we just defined, and by d the exterior derivative.

First we claim that $\nabla_X(\omega \wedge \eta) = \nabla_X(\omega) \wedge \eta + \omega \wedge \nabla_X \eta$ for all forms ω, η and $X \in \mathfrak{X}(M)$. This is because $\nabla_X^\sigma F = \sigma \nabla_X F$ for all covariant tensors F and permutations σ . This is immediate from

$$\nabla_X F(\bar{Y}) = X(F(\bar{Y})) - \sum_i F(Y_1, \dots, \nabla_X Y_i, \dots, Y_n).$$

Then ∇_X commutes with Alt since it is just a linear combination of σ s. So

$$\begin{aligned} \nabla_X(\omega \wedge \eta) &= \nabla_X \text{Alt}(\omega \otimes \eta) = \text{Alt} \nabla_X(\omega \otimes \eta) = \text{Alt}((\nabla_X \omega) \otimes \eta) + \text{Alt}(\omega \otimes \nabla_X \eta) \\ &= (\nabla_X \omega) \wedge \eta + \omega \wedge \nabla_X \eta. \end{aligned}$$

Now we claim that $D(dx^I)$ for any multi-index I of length k . It is sufficient to show that for any multi-index J of length $k+1$, $D(dx^I)(\partial_J) = 0$, where $\partial_J = (\partial_{j_1}, \dots, \partial_{j_{k+1}})$. For a multi-index $I = (i)$ then we have

$$D(dx^i)(\partial_j, \partial_\ell) = \nabla_{\partial_j} dx^i(\partial_\ell) - \nabla_{\partial_\ell} dx^i(\partial_j).$$

We know that for 1-forms,

$$\nabla_{\partial_j} dx^i(\partial_\ell) = \partial_\ell(dx^i(\partial_j)) - dx^i(\nabla_{\partial_j} \partial_\ell) = -dx^i(\nabla_{\partial_j} \partial_\ell),$$

since $dx^i \partial_j$ is constant. So this is equal to

$$D(dx^i)(\partial_j, \partial_\ell) = dx^i(\nabla_{\partial_\ell} \partial_j - \nabla_{\partial_j} \partial_\ell) = dx^i[\partial_\ell, \partial_j] = 0.$$

So we have our base case.

Now consider the multi-index (I, ℓ) , where I has length k . Writing $\widehat{J}_i$ for the multi-index obtained by removing the index i from J , we have

$$D(dx^I \wedge dx^\ell)(\partial_J) = \sum_i (-1)^{i+1} \nabla_{\partial_{j_i}} dx^I \wedge dx^\ell(\partial_{\widehat{J}_i}) + \sum_i (-1)^{i+1} dx^I \wedge \nabla_{\partial_{j_i}} dx^\ell(\partial_{\widehat{J}_i}).$$

We claim that both sums here are zero.

For the first sum, if $\ell \notin J$ then it is trivially zero. Otherwise, we can assume that it is the last index in J (because we can otherwise permute J and this will affect the sum only in sign), so let $J = (J', \ell)$. Then the first sum is

$$\sum_{i \leq k+1} (-1)^{i+1} \nabla_{\partial_{j_i}} dx^I(\partial_{\widehat{J}'_i}) = D(dx^I)(\partial_{J'}) = 0$$

inductively.

For the second sum, if I is not a sub-multi-index of J then this is trivially zero again. Otherwise we can assume I occurs as the first k indices of J , $J = (I, \partial_j, \partial_m)$ (otherwise we permute). Then in the sum, we can take i to be greater than k as otherwise we remove an index from I in J and the resulting wedge will be zero. So the sum is then equal to

$$\pm(\nabla_{\partial_j} dx^\ell(\partial_m) - \nabla_{\partial_m} dx^\ell(\partial_j)) = \pm D(dx^\ell)(\partial_j, \partial_m) = 0$$

from our base case. So both sums are zero and thus inductively we have $D(dx^I) = 0$.

By linearity of D it is sufficient to show that $D(udx^I) = du \wedge dx^I$ for $u \in C^\infty(M)$. Indeed, from what we just showed:

$$\begin{aligned} D(udx^I)(\partial_J) &= \sum_i (-1)^{i+1} \nabla_{\partial_{j_i}} u dx^I(\partial_{\widehat{J}_i}) \\ &= \sum_i (-1)^{i+1} (\partial_{j_i} u) dx^I(\partial_{\widehat{J}_i}) + u \sum_i (-1)^{i+1} \partial_{j_i} dx^I(\partial_{\widehat{J}_i}) \\ &= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^I + u D(dx^I)(\partial_J) \\ &= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^I. \end{aligned}$$

On the other hand,

$$du \wedge dx^I(\partial_J) = \sum_i \frac{\partial u}{\partial x_i} dx^i \wedge dx^I(\partial_J),$$

and we can sum over i occuring in J as otherwise the term will be zero,

$$= \sum_i \frac{\partial u}{\partial x_{j_i}} dx^{j_i} \wedge dx^I(\partial_J) = \sum_i \frac{\partial u}{\partial x_{j_i}} \delta_J^{j_i I}.$$

Since j_i occurs at the i th index of J , a permutation of J bring it to $j_i I$ must map i to 1 and gives a permutation of $\widehat{J}_i$ mapping to I . The sign of this permutation is $(-1)^{i+1}$ times the sign of the sub-permutation mapping $\widehat{J}_i$ to I , as we must offset all the indices before i by 1. So this is equal to

$$= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{J_i}^I = D(u dx^I),$$

as desired.

It is sufficient to show this equality for ∂_J since they are both alternating and ∂_J forms local frames.

(4) The formula

$$d\omega = \sum_I d\omega_I \wedge dx^I$$

is independent on choice of metric and connection, and it coincides with the construction from class.

(5) We know that

$$\nabla_Y \omega(\overline{X}) = Y(\omega(\overline{X})) - \sum_i \omega(X_1, \dots, \nabla_Y X_i, \dots, X_k).$$

Thus

$$\begin{aligned} d\omega(X_1, \dots, X_{k+1}) &= \sum_i (-1)^{i+1} \nabla_{X_i} \omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1}) \\ &= \sum_i (-1)^{i+1} X_i(\omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1})) - \sum_i (-1)^{i+1} \sum_{j \neq i} \omega(\dots, \widehat{X}_i, \dots, \nabla_{X_i} X_j, \dots). \end{aligned}$$

In the final sum, $\widehat{X}_i$ may come after $\nabla_{X_i} X_j$. This final sum can thus be written as

$$\sum_{i < j} (-1)^{j+1} \omega(\dots, \nabla_{X_j} X_i, \dots, \widehat{X}_j, \dots) + \sum_{i < j} (-1)^{i+1} \omega(\dots, \widehat{X}_i, \dots, \nabla_{X_i} X_j, \dots).$$

Permuting the inputs so that the covariant derivative appears first has a sign of $(-1)^{i+1}$ in the first sum and $(-1)^j$ in the second ($i-1$ and j transpositions respectively). So this is equal to

$$= \sum_{i < j} (-1)^{i+j} (\omega(\nabla_{X_j} X_i, \dots) - \omega(\nabla_{X_i} X_j, \dots)),$$

and by symmetry of the connection, this is equal to

$$= \sum_{i < j} (-1)^{i+j} \omega([X_j, X_i], \dots) = - \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], \dots).$$

We subtract this from the first sum, so we see that

$$d\omega(X_1, \dots, X_{k+1}) = \sum_i (-1)^{i+1} X_i(\omega(\dots \widehat{X}_i \dots)) + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], \dots)$$

as desired.